ArtConnect Pro

Database Architecture & Java Implementation

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OVERVIEW

What is ArtConnect Pro?

Art Community

Manage artists, artworks, galleries, exhibitions, workshops & community members in one platform.

Relational DB

MySQL database with 13 tables, foreign keys, views, triggers, and stored procedures.

Java Layer

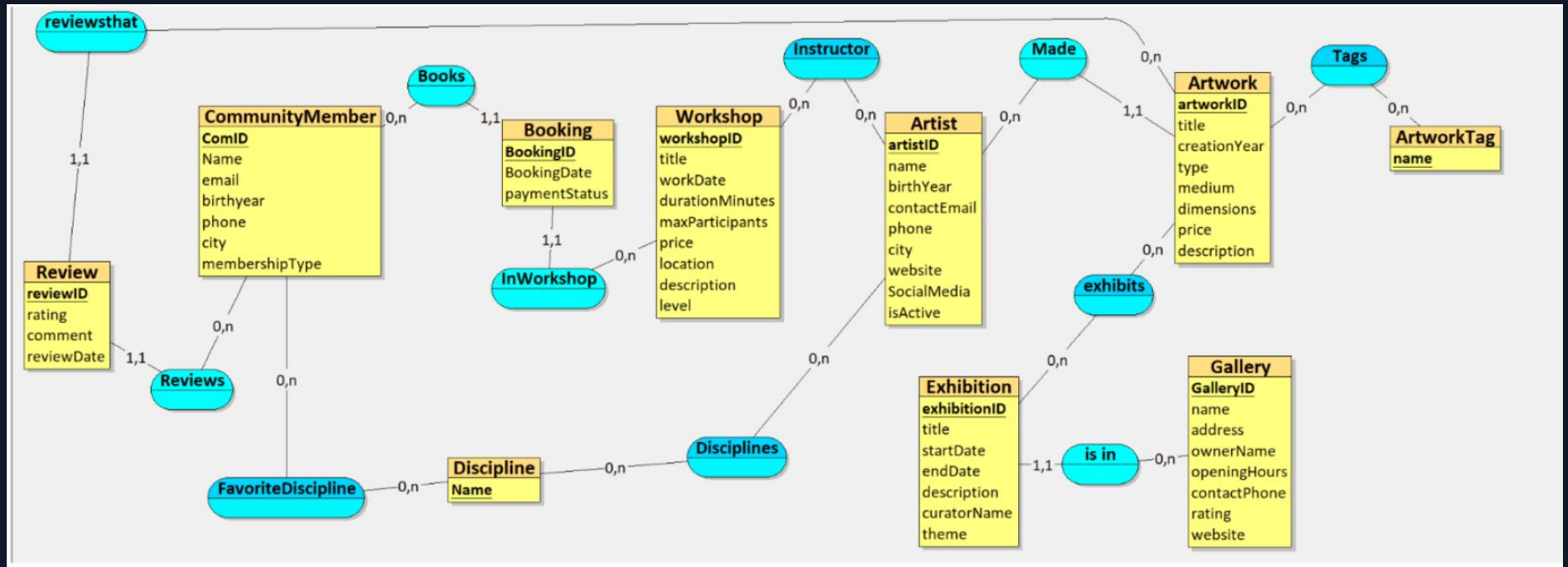
JavaFX UI + Service layer + DAO interfaces backed by JDBC implementations connecting to MySQL.

STEP 2 · SCHEMA

Table Relational Schema

Artist	Core entity – name, city, disciplines, isActive
Artwork	Linked to Artist via artistID (FK)
ArtworkTag	Tag vocabulary (abstract, urban…)
Tags	Artwork ↔ ArtworkTag (M:N)
Gallery	Venue – address, hours, rating
Exhibition	Linked to Gallery via GalleryID (FK)
exhibits	Artwork ↔ Exhibition (M:N)
Workshop	Title, date, capacity, level, price
Instructor	Artist ↔ Workshop (M:N)
CommunityMember	Name, city, membershipType
Booking	Member ↔ Workshop + paymentStatus
Review	Member reviews an Artwork (rating 1-5)
FavoriteDiscipline	Member ↔ Discipline (M:N)

Entity Relationships



STEP 3 · VIEWS

3 SQL Views

PublicArtistProfile

Objective: Security

Hides phone and personal email from end-users. Shows name, city, disciplines and public links only. Internal queries go directly to the Artist table (admin-restricted).

ExhibitionCatalogue

Objective: Simplification

Pre-joins Exhibition → Gallery → exhibits → Artwork → Artist → Tags. Turns a 6-table query into a simple SELECT, ideal for the catalogue screen.

ActiveExhibitions

Objective: Security + Temporal

Filters WHERE startDate ≤ TODAY ≤ endDate. Encapsulates time logic in one place.

STEP 3 · TRIGGERS

3 Data-Integrity Triggers

trg_check_workshop_capacity

BEFORE INSERT on Booking

Counts existing bookings for the workshop. If current_count ≥ maxParticipants → raises SQLSTATE 45000 'Workshop is fully booked'.

trg_check_exhibition_dates

BEFORE INSERT / UPDATE on Exhibition

Ensures endDate ≥ startDate. Two mirrored triggers cover both INSERT and UPDATE to prevent invalid date ranges from ever reaching the table.

trg_check_booking_date

BEFORE INSERT on Booking

Fetches the workshop's workDate and compares with CURDATE(). Prevents members from booking workshops that have already taken place.

STEP 3 · FUNCTIONS & PROCEDURES

Functions & Stored Procedures

Functions

fn_get_artwork_avg_rating(artwork)

→ DECIMAL(4,2)

Average review rating for an artwork (0.00 if no reviews).

fn_get_participant_count(workshop)

→ INT

Count of Paid bookings for a workshop.

fn_is_member_booked(workshop,name)

→ BOOLEAN

Returns TRUE if member already has a booking for the workshop.

Stored Procedures

sp_create_workshop_with_instructor

Validates artist is active, then wraps Workshop INSERT + Instructor INSERT in a TRANSACTION.

sp_add_artist_with_discipline

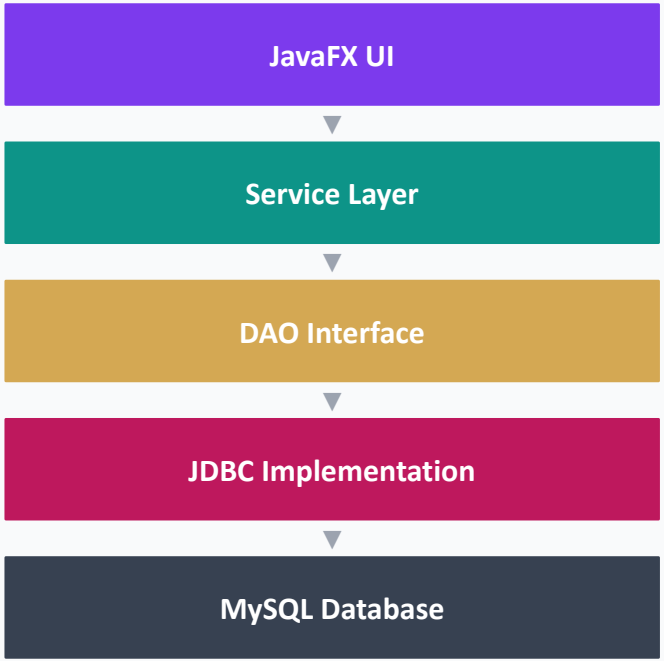
Inserts a new Artist and auto-creates the Discipline if it doesn't exist yet.

sp_workshop_report

Returns 3 result sets: workshop details, instructor list, and registered members.

JAVA IMPLEMENTATION

JDBC DAO Layer



Key Design Decisions

OOP-First, No IDs in Models

Java objects use direct references (Artwork → Artist). JDBC layer resolves DB primary keys by name at query time.

Modular Design

The use of interfaces makes the switching of memory type (database / in memory) easy. The integration and implementation is incremental.

API design

The whole library exposes a single API.

Singleton Connection

ConnectionManager lazy-initialises one shared Connection; re-creates it if closed. closeConnection() called on shutdown.